



3.5 Trigonometric identities and equations

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Key identities: Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$ (and derived $\tan^2\theta + 1 = \sec^2\theta$); double angle formulas $\sin(2\theta) = 2\sin\theta\cos\theta$, $\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$; $\tan\theta = \sin\theta/\cos\theta$. To solve trig equations over a given domain, use identities to reduce to a single function, find the principal value, then use the unit circle/graph symmetry to find all solutions in range.

Worked example: follow the reasoning

Solve $2\cos^2x - 1 = 0$ for $0 \leq x \leq 2\pi$.

$\cos^2x = 1/2$, $\cos x = \pm 1/\sqrt{2}$.

$x = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4$ (all four quadrant solutions where cosine is $\pm 1/\sqrt{2}$).

Quick check · multiple choice

Simplify $\sin(2x) / (2\sin x)$ for $\sin x \neq 0$.

- A $\cos x$
- B $\sin x$
- C $2\cos x$
- D $\tan x$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 (a) Show that the equation $\cos(2x) + 3\cos x - 1 = 0$ can be written as $2\cos^2x + 3\cos x - 2 = 0$. [7]
- (b) Hence solve $\cos(2x) + 3\cos x - 1 = 0$ for $0 \leq x \leq 2\pi$, giving exact or 3 s.f. answers.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $\cos(2x) = 2\cos^2x - 1$, so equation becomes $2\cos^2x - 1 + 3\cos x - 1 = 0 \rightarrow 2\cos^2x + 3\cos x - 2 = 0$.

(b) Let $c = \cos x$: $2c^2 + 3c - 2 = 0 \rightarrow (2c - 1)(c + 2) = 0 \rightarrow c = 1/2$ or $c = -2$ (rejected, outside $[-1, 1]$).
 $\cos x = 1/2 \rightarrow x = \pi/3$ or $x = 2\pi - \pi/3 = 5\pi/3$ (in $[0, 2\pi]$).

Quick check answer

A. $\sin(2x) = 2\sin x \cos x$, so $\sin(2x)/(2\sin x) = 2\sin x \cos x/(2\sin x) = \cos x$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.